

Karatsuba Fast Multiplication Algorithm

Let $x = x_2 \cdot 10^N + x_1$, $y = y_2 \cdot 10^N + y_1$, where $x_i, y_i \in I(N)$ for $i \in \{1, 2\}$

Hence, $z = x \cdot y = x_2 \cdot y_2 \cdot 10^{2N} + (x_2 \cdot y_1 + x_1 \cdot y_2) \cdot 10^N + x_1 \cdot y_1$ with time complexity $O(N^2)$

Let $a = x_2 \cdot y_2$, $b = x_1 \cdot y_1$, $c = (x_2 + x_1) \cdot (y_2 + y_1)$

Hence, $x_2 \cdot y_1 + x_1 \cdot y_2 = (x_2 + x_1) \cdot (y_2 + y_1) - x_1 \cdot y_1 - x_2 \cdot y_2 = c - b - a$

Hence, $z = x \cdot y = a \cdot 10^{2N} + (c - b - a) \cdot 10^N + b$ with time complexity $O(N^{\log_2 3}) \approx O(N^{1.585})$